

TABLE 7—NONLINEAR MODELS OF EVACUEE SHARE IN GRADE AND EVACUEE ACHIEVEMENT ON INCUMBENT ACHIEVEMENT: LOUISIANA MIDDLE/HIGH

|                                  | All                | Bottom            | 2nd                | 3rd                | Top                |
|----------------------------------|--------------------|-------------------|--------------------|--------------------|--------------------|
|                                  | (1)                | quartile          | quartile           | quartile           | quartile           |
| Pre-Katrina LEAP quartile        | (2)                | (3)               | (4)                | (5)                |                    |
| <i>Panel A. Math</i>             |                    |                   |                    |                    |                    |
| Katrina/Rita share in quartile 1 | −0.79***<br>(0.25) | −0.68**<br>(0.31) | −1.56***<br>(0.29) | −1.31***<br>(0.38) | −1.67***<br>(0.54) |
| Katrina/Rita share in quartile 2 | −0.23<br>(0.31)    | −0.56<br>(0.45)   | −0.41<br>(0.40)    | −0.11<br>(0.38)    | 0.02<br>(0.61)     |
| Katrina/Rita share in quartile 3 | 0.34<br>(0.31)     | 0.32<br>(0.47)    | 0.69*<br>(0.40)    | 0.73*<br>(0.39)    | 0.48<br>(0.51)     |
| Katrina/Rita share in quartile 4 | 0.42<br>(0.31)     | 1.25**<br>(0.54)  | 1.24***<br>(0.41)  | 0.75*<br>(0.39)    | 0.93*<br>(0.56)    |
| Observations                     | 536,009            | 123,316           | 133,757            | 137,800            | 141,136            |
| Estimate of Q4–Q1                | 1.21***<br>(0.44)  | 1.93***<br>(0.66) | 2.79***<br>(0.54)  | 2.06***<br>(0.60)  | 2.59***<br>(0.83)  |
| <i>Panel B. ELA</i>              |                    |                   |                    |                    |                    |
| Katrina/Rita share in quartile 1 | −0.16<br>(0.28)    | −0.44<br>(0.35)   | −0.45<br>(0.38)    | −1.02***<br>(0.35) | −0.53<br>(0.44)    |
| Katrina/Rita share in quartile 2 | 0.03<br>(0.35)     | 0.74<br>(0.57)    | −0.18<br>(0.48)    | −0.40<br>(0.44)    | −0.51<br>(0.42)    |
| Katrina/Rita share in quartile 3 | −0.23<br>(0.35)    | 0.18<br>(0.58)    | −0.03<br>(0.48)    | −0.13<br>(0.42)    | −0.03<br>(0.47)    |
| Katrina/Rita share in quartile 4 | 0.37<br>(0.27)     | 1.62**<br>(0.65)  | 1.11**<br>(0.45)   | 1.81***<br>(0.36)  | 0.46<br>(0.40)     |
| Observations                     | 522,392            | 116,304           | 128,894            | 135,911            | 141,283            |
| Estimate of Q4–Q1                | 0.54<br>(0.42)     | 2.06***<br>(0.77) | 1.56**<br>(0.66)   | 2.83***<br>(0.53)  | 0.98<br>(0.64)     |

*Notes:* Standard errors are provided in parentheses and clustered by school. LEAP scores are standard deviations of scale scores within grade and year. The standardization uses nonevacuees in schools where there are < 70 percent evacuees and excludes schools in Orleans, Jefferson, St. Bernard, Plaquemines, Cameron, and Calcasieu parishes. When students have multiple scores for a single subject in a given year we use the lowest score. Quartiles are determined using the most recent pre-Katrina score available and are within grade and year. Regressions also include controls for gender, race, free/reduced-price lunch eligibility, school fixed-effects, grade-by-year fixed-effects, and, for middle/high, lagged achievement from the student’s most recent pre-Katrina lagged score interacted with indicators for the number of years between exams. Elementary includes grade 4 from 1999–2000 through 2004–2005 and grade 5 in 2005–2006. Middle/High includes grade 10 in 2001–2002 and 2002–2003, grades 8 and 10 in 2003–2004 and 2004–2005, and grades 6–10 in 2005–2006 and 2006–2007.

\*\*\*Significant at the 1 percent level.  
\*\*Significant at the 5 percent level.  
\*Significant at the 10 percent level.

top-quartile evacuees are large and significant for math and ELA. For example, a 10 percentage point increase in bottom-quartile evacuees reduces math achievement of top-quartile incumbents by 0.17 standard deviations while the impact on bottom-quartile incumbents is smaller at 0.07 standard deviations. On the other hand, a similar increase in top-quartile evacuees increases incumbent scores by 0.13 in the bottom quartile and by 0.09 in the top quartile. The impact on ELA is similar; a 10 percentage point increase in bottom quartile evacuees reduces incumbent achievement by 0.04 to 0.10 standard deviations. A similar increase in top-quartile evacuees increases incumbent achievement by 0.05 to 0.18 standard deviations. Given that a 10 percentage point shift at the top or bottom of the achievement distribution is a large change in peer composition, these estimates are